

Aho-Corasick Algorithm – Research & Explanation

The Aho-Corasick Algorithm is an efficient string matching algorithm used to search multiple patterns simultaneously within a large text. It combines the Trie data structure with failure links to achieve fast searching.

1. What is Aho-Corasick?

Aho-Corasick is a multi-pattern string matching algorithm developed by Alfred V. Aho and Margaret J. Corasick in 1975. Unlike normal searching methods that search one word at a time, this algorithm can search many keywords together in a single pass.

The algorithm is mainly based on:

- Trie Data Structure
- Failure Links
- Efficient Pattern Matching

2. How Does It Work?

Step 1: Build a Trie

All patterns are inserted into a Trie. Each node represents a character and paths form complete words.

Step 2: Create Failure Links

Failure links help the algorithm avoid restarting the search from the beginning. If a mismatch occurs, the algorithm jumps to the longest possible suffix that is also a prefix of another pattern.

Step 3: Search the Text

The text is scanned character by character. Whenever a pattern is matched, the algorithm reports it immediately.

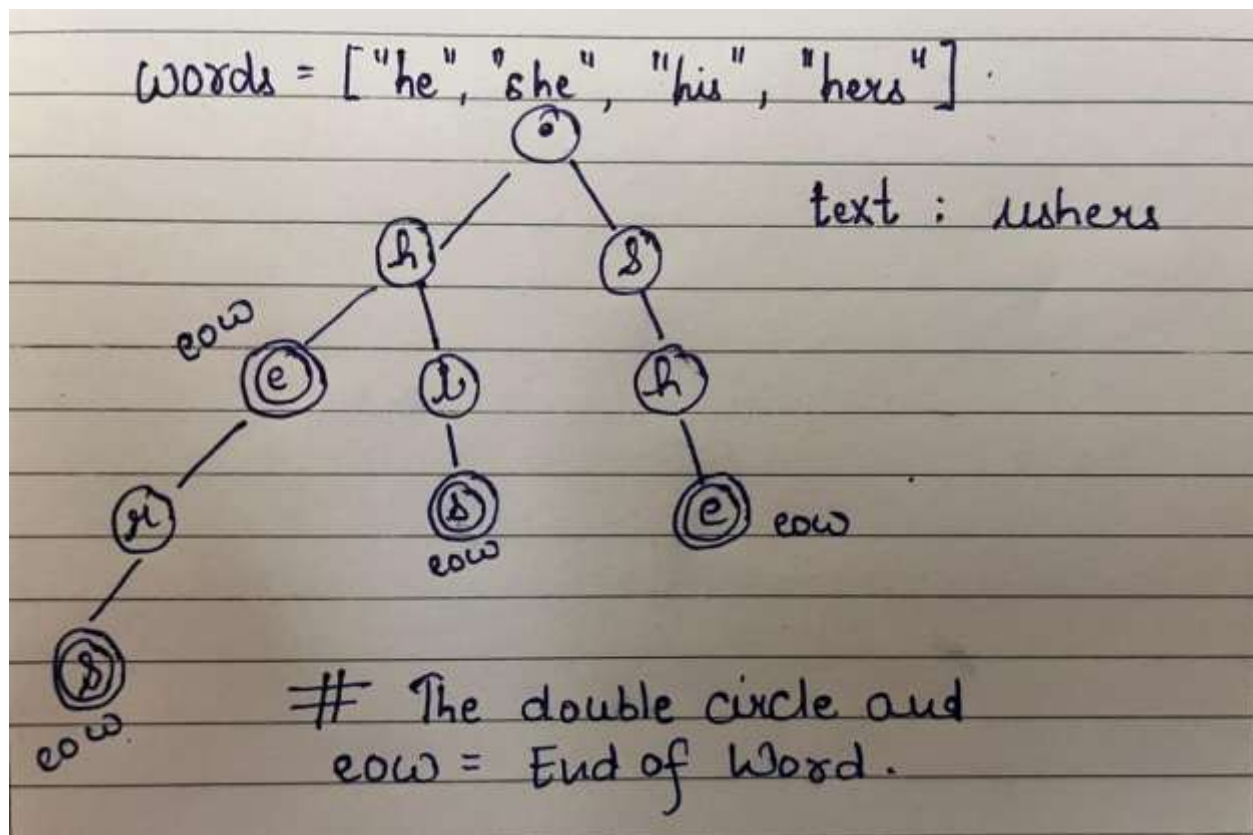
Example:

Patterns = {he, she, his, hers}

Text = 'ushers'

Matches Found:

- she
- he
- hers



3. Why is it Efficient?

Aho-Corasick is efficient because it processes the text only once. Instead of checking each pattern separately, it searches all patterns together.

Advantages:

- Faster than naive searching
- Handles multiple patterns together
- Avoids unnecessary comparisons using failure links
- Suitable for large datasets

4. Real-World Applications

- Search engines for keyword detection
- Spell checking systems
- Antivirus software for malware signature detection
- DNA sequence matching in bioinformatics
- Spam filtering systems
- Intrusion detection systems in cybersecurity
- Text editors and document search tools

5. Time and Space Complexity

Operation	Complexity
Building the Trie	$O(L)$
Creating Failure Links	$O(L)$
Searching the Text	$O(L + M)$
Space Complexity	$O(\text{total number of Trie nodes})$

6. Conclusion

The Aho-Corasick Algorithm is one of the most efficient algorithms for multi-pattern string matching. Its combination of Trie structure and failure links allows fast searching with minimal repeated work, making it highly useful in real-world applications such as cybersecurity, search engines, and bioinformatics.